CDI COLLEGE | BDA400

Assignment 6

Technical Analysis Using R

Visualization Stage | Portfolio Dashboard

Student and Course

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Course	CA-BDA400 - Data Science Tools and Techniques
Assignment	Assignment 6: Technical Analysis Using R, Visualization Stage

Project Overview

This project builds an interactive R Shiny portfolio dashboard using historical stock data from Yahoo Finance. It lets users examine NVDA, AMZN, and TSLA over a selected date range and at daily, weekly, or monthly intervals. Price data can be viewed as line, area, or candlestick charts.

Dashboard Features

- Interactive stock, date-range, time-frame, and chart-type controls.
- Optional 20-period and 50-period moving averages, RSI (14), and MACD indicators.
- A short/long moving-average crossover rule that identifies Buy, Sell, or Hold states and labels Buy/Sell crossover points on the price chart.

Public Assignment Repository

https://github.com/lubnaafandi89-coder/TechnicalAnalysis/tree/main/Assignment6

The repository contains the Shiny application source code and project notes.